OV7725_D[7:0] OV7725_HREF OV7725_PCLK OV7725_VSYNC

rst

SwitchUart

sw_ColorClear

button_eeprom

btn_ColorExrtract btn_ColorExrtract1
btn_ColorExrtract2

clk100

debounce_0

debounce

clki o

cam_ov7670_ov7725_0

cam_ov7670_ov7725

href pclk
vsync wclk we

d[7:0] addr[16:0] dout[15:0]

vga_0

vga

vga_pclk vga_hsync

vga_rst vga_valid
vga_vsync

vga_h_cnt[11:0]

vga_v_cnt[10:0]

blk_mem_gen_0

blk_mem_gen_0

clka
clkb

addra[16:0] addrb[16:0]

dina[15:0]

doutb[15:0]

wea

region_cut_0

region_cut

valid_h_cnt[11:0] valid_v_cnt[10:0] vga_h_cnt[11:0] vga_v_cnt[10:0]

erode_0

erode

PCLK

pix_i

VtcHCnt[11:0] pix_o VtcVCnt[10:0]

erode_1

erode

PCLK

pix_i

VtcHCnt[11:0] pix_o VtcVCnt[10:0]

erode_2

erode

PCLK

pix_i

VtcHCnt[11:0] pix_o VtcVCnt[10:0]

rgb565_rgb888_0

rgb565_rgb888

rgb565[15:0] rgb888[23:0]

rst_n0_i__0 RTL_INV

I0 O

dilate_0

dilate

PCLK

pix_i

VtcHCnt[11:0] pix_o VtcVCnt[10:0]

dilate_1

dilate

PCLK

pix_i

VtcHCnt[11:0] pix_o VtcVCnt[10:0]

dilate_2

dilate

PCLK

pix_i

VtcHCnt[11:0] pix_o VtcVCnt[10:0]

rgb2hsv_top_0

rgb2hsv_top

pclk RGB24[23:0] HSV24[23:0]

u_e2prom_top

top_e2prom

button_eeprom

iic_scl
iic_sda
led

on1Switch
on2Switch
on3Switch
sys_clk read_flag
sys_rst_n

data_on1Switch[23:0] data_on2Switch[23:0] data_on3Switch[23:0]

indata_on1Switch[23:0]
indata_on2Switch[23:0]
indata_on3Switch[23:0]

rst_n0_i RTL_INV

I0 O

colorDetect_0

colorDetect

Binary_PostProcess Binary_PostProcess1 Binary_PostProcess2 Binary_PreProcess Binary_PreProcess1 PClk Binary_PreProcess2
btn_ColorExtract btn_ColorExtract1
btn_ColorExtract2

read_flag
rst_n
sw_ColorClear

HSV24[23:0] RGB24[23:0] RGB24_dis[23:0] VtcHCnt[11:0] VtcVCnt[10:0] center_h[11:0] center_h1[11:0] center_h2[11:0] center_v[10:0] center_v1[10:0] data_on1Switch[23:0] center_v2[10:0] data_on2Switch[23:0] data_on3Switch[23:0]

indata_on1Switch[23:0]
indata_on2Switch[23:0]
indata_on3Switch[23:0]

angle_calculation

angle_calculation

clk
rst_n

angle[7:0] site_x1[15:0] site_x2[15:0] site_x3[15:0] site_y1[15:0] site_y2[15:0] site_y3[15:0]

uarttemp_i

RTL_MUX S

S=1'b1 I0[7:0] S=default I1[7:0] O[7:0]

ov7725_regData_0

ov7725_regData

LUT_INDEX[7:0] LUT_DATA[15:0] Slave_Addr[7:0] V=B"01000010"

rgb888_rgb565_0

rgb888_rgb565

rgb888[23:0] rgb565[15:0]

hex3_in_i

RTL_MUX S

V=B"1010", S=1'b1 I0[3:0] V=B"1100", S=default I1[3:0] O[3:0]

clk_wiz_0

clk_wiz_0

clk_in1 clk_out1

bin2bcd_0

bin2bcd

bin[7:0] bcd[11:0]

clk_uart38400

clk_div

clk clk_out

IICctrl_0

IICctrl

I2C_SCLK
I2C_SDAT
iCLK
rst

LUT_DATA[15:0]

LUT_INDEX[7:0] V=B"01000010" Slave_Addr[7:0]

ram_read_0

ram_read

clk

valid

address[16:0] din[15:0] vga_blue[3:0] vga_green[3:0] vga_h_cnt[11:0] vga_red[3:0] vga_v_cnt[10:0]

data_display_led_i

RTL_MUX S

S=1'b1 I0[7:0] S=default I1[11:0] O[11:0]

scan_led_hex_disp_0

scan_led_hex_disp

clk

reset

an[3:0] dp_in[3:0] hex0[3:0] hex1[3:0] hex2[3:0] hex3[3:0]

sseg[7:0]

uart_tx_0

uart_tx

clk txd
data_o[7:0]

OV7725_SIOC OV7725_SIOD

vga_blue[3:0] vga_green[3:0] vga_red[3:0]
vga_hsync

vga_vsync OV7725_XCLK

data_display_led[11:0]

an[3:0] seg[7:0]

sdat

led

txd_ov7725

sclk